Problem 5: Super Ben Galaxy
7+2 Points

Problem ID: `galaxy`
Rank: `2+3`

Introduction

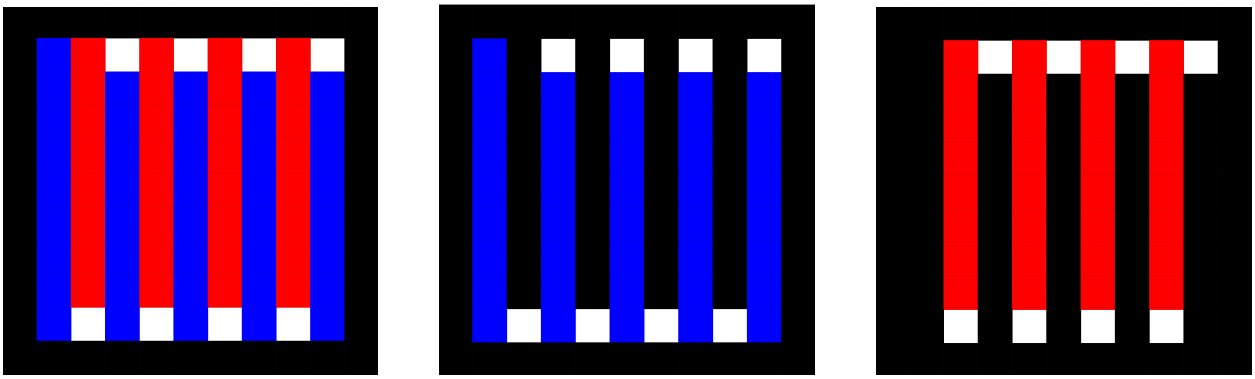
Every hundred years, a lightbulb appears in the skies of UC Berkeley. The lightbulb was so large one year, it filled the skies and sent paint raining down. The CALICO team brought the paint to the castle, where they became the great Paint Bucket. It should have been a very happy time for the citizens of UC Berkeley. That was the night of the CALICO Contest, held once every semester to celebrate the lightbulb.



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Problem Statement

You are given an $\mathbf{N} \times \mathbf{M}$ grid $\mathbf{G}$, consisting of red (`R`), blue (`B`) and normal (`.`) tiles. You are only able to stand or walk onto tiles which are active. Normal tiles are always active, red tiles are active only in red tile mode, and blue tiles are active only in blue tile mode. If you are standing on a normal tile, you may choose to perform a spin action, which toggles which mode you are in (red tile mode to blue tile mode or vice versa). See Figure 1 below for a visualization of the third main sample test case, with white representing normal tiles and black representing tiles that cannot be walked on:



(a) Full Map (b) Blue Tile Mode (c) Red Tile Mode

Figure 1: Visualizations of Main Sample Test Case 3

You are given $\mathbf{Q}$ queries, the ith of which contains Ben's starting location at $(\mathbf{R_{start}}[i], \mathbf{C_{start}}[i])$ and Ben's intended destination at $(\mathbf{R_{end}}[i], \mathbf{C_{end}}[i])$. You may assume that the tile at Ben's starting location is normal, and that we start in blue tile mode.

Main Test Set

For the main test set, it is guaranteed that $\mathbf{R_{end}}[i] = \mathbf{N} - 1$ and $\mathbf{C_{end}}[i] = \mathbf{M} - 1$ for every query i.

For each query, return whether Ben can successfully get from his starting location at $(\mathbf{R_{start}}[i], \mathbf{C_{start}}[i])$ to his destination, which for this test set will always be at the bottom right corner $(\mathbf{N} - 1, \mathbf{M} - 1)$.

Bonus Test Set

For the bonus version, it is **no longer guaranteed that Ben's destination is the bottom right corner**. That is, for each query, return whether Ben can successfully get from his starting location at $(\mathbf{R_{start}}[i], \mathbf{C_{start}}[i])$ to his destination at $(\mathbf{R_{end}}[i], \mathbf{C_{end}}[i])$.

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Input Format

The first line of the input contains a single integer $\mathbf{T}$ denoting the number of test cases that follow. For each test case:

- The first line contains three space-separated integers $\mathbf{N}$ $\mathbf{M}$ $\mathbf{Q}$, where $\mathbf{N}$ is the number of rows, $\mathbf{M}$ is the number of columns, and $\mathbf{Q}$ is the number of queries.
- Each of the next $\mathbf{N}$ lines contains a string of length $\mathbf{M}$ representing a row in the grid. Each character in the string is either `R`, `B` or `.`
- Each of the next $\mathbf{Q}$ lines is a query containing space-separated integers $\mathbf{R_{start}}[i]$ $\mathbf{C_{start}}[i]$ $\mathbf{R_{end}}[i]$ $\mathbf{C_{end}}[i]$ in that order, representing the zero-indexed coordinates of Ben's starting location and destination for the $\mathbf{i}$th query. Note that for the main test set, it is guaranteed that $\mathbf{R_{end}}[i] = \mathbf{N} - 1$ and $\mathbf{C_{end}}[i] = \mathbf{M} - 1$.

Output Format

For each test case, output $\mathbf{Q}$ lines. Each line should either be `YES` if it is possible for Ben to go from his starting location to his destination for the corresponding query, or `NO` if not.

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Constraints

$1 \leq \mathbf{T} \leq 100$
$1 \leq \mathbf{N} \leq 200$
$1 \leq \mathbf{M} \leq 200$
$1 \leq \mathbf{Q} \leq 10^5$
$0 \leq \mathbf{R_{start}}[i] < \mathbf{N}$ for all i
$0 \leq \mathbf{C_{start}}[i] < \mathbf{N}$ for all i
The tile at $(\mathbf{R_{start}}[i], \mathbf{C_{start}}[i])$ is normal.
The sum of $\mathbf{N} \times \mathbf{M}$ over all test cases is at most 2×10^5.
The sum of $\mathbf{Q}$ over all test cases is at most 10^5.

Main Test Set

$\mathbf{R_{end}}[i] = \mathbf{N} - 1$ for all i
$\mathbf{C_{end}}[i] = \mathbf{M} - 1$ for all i

Bonus Test Set

$0 \leq \mathbf{R_{end}}[i] < \mathbf{N}$ for all i
$0 \leq \mathbf{C_{end}}[i] < \mathbf{M}$ for all i

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Sample Test Cases

Main Test Set

Sample Input Download

```
3
2 4 2
.R.B
.RRB
0 0 1 3
1 0 1 3
2 4 2
.RB.
.RB.
0 0 1 3
0 3 1 3
9 9 3
BR.R.R.R.
BRBRBRBRB
BRBRBRBRB
BRBRBRBRB
BRBRBRBRB
BRBRBRBRB
BRBRBRBRB
BRBRBRBRB
B.B.B.B.B
8 1 8 8
0 4 8 8
8 7 8 8
```

Sample Output Download

```
YES
YES
NO
YES
YES
YES
YES
```

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Bonus Test Set

Sample Input [Download]

```
2
2 4 2
.RB.
.RB.
1 0 1 2
1 3 0 3
9 9 3
BR.R.R.R.
BRBRBRBRB
BRBRBRBRB
BRBRBRBRB
BRBRBRBRB
BRBRBRBRB
BRBRBRBRB
BRBRBRBRB
B.B.B.B.B
0 8 4 5
8 1 2 8
8 7 0 0
```

Sample Output [Download]

```
NO
YES
YES
YES
YES
```

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Sample Explanations

For the following diagrams, green tiles represent the current tile that Ben is standing on.

Figure 2 shows the full map for the first main sample test case, with both blue and red tiles active for the purposes of visualization. Note that in the actual problem, blue and red tiles will never be active at the same time.

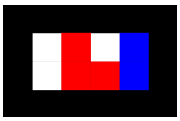


Figure 2: Full Map

For the first test case's first query, Figure 3 displays the initial state of the grid in blue mode, and Ben is at his starting location in the top left corner.

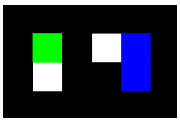


Figure 3: Initial State

Ben then performs a spin action while at the top left corner, switching the state of the grid to red mode in Figure 4, and Ben is able to move to the cell at $(0, 2)$.

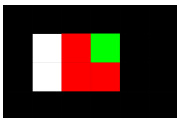


Figure 4: First Spin

Ben uses a second spin action at this cell to switch the grid back to blue mode, and he is able to reach the bottom right corner, as shown in Figure 5.

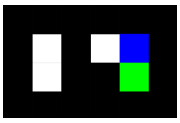


Figure 5: Second Spin

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